DO NOW — RECALL

Year 13 autumn mock: Paper 1

Name: _____ Class: _____ Date: _____

1. Evaluate true AND false.

2. Why should a model omit irrelevant details?

3. What does the Program Counter hold?

4. A full adder receives A=1, B=1 and carry-in=1. State sum and carry-out and justify using binary addition.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

False.

2. Model answer

To reduce complexity while retaining data needed to answer the question.

3. Model answer

The address of the next instruction to fetch.

4. Model answer

1+1+1=3 decimal=11 binary; Sum=1, carry-out=1.

Vocabulary — know these terms

diagnostic — Assessment used to find knowledge gaps.

two's complement — Encoding signed binary integers.

De Morgan's law — Negate operands and swap AND/OR.

register — Fast CPU storage for a specific role.

NEW LEARNING

Read, annotate and explain

Purpose

This is an original diagnostic Paper 1 practice, not an official OCR examination paper. Some systems topics will be retaught later.

Manage time

Allow approximately 15 minutes to each 15-mark question.

Show methods

Write the inversion/add-one steps, state every Boolean simplification rule and use precise CPU register terminology.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Unmarked example: what earns method marks for a negative 8-bit value?

Predict the method

Write +5 in exactly 8 bits.

Check and explain

Invert each bit, then add one to obtain the two's complement result for −5.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

(a) Convert −18 to eight-bit two's complement, showing working. [4] (b) Perform arithmetic right shift by 1 on that representation, giving the binary result and signed decimal value. [3] (c) Add 1011_2 and 0110_2, showing carry and answer. [3] (d) Normalise positive $0.0011_2 \times 2^5$, showing shifts and decimal value. [5]

Question 2

(a) Complete four output rows of A XOR B in order 00,01,10,11. [4] (b) Simplify ¬(A ∧ B) using De Morgan's law. [3] (c) For a half adder with A=1, B=1, state Sum and Carry and explain the gates. [4] (d) Simplify A ∨ (B ∧ ¬B) using Boolean identities. [4]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

(a) State the roles of PC, MAR, MDR, CIR and ACC. [5] (b) Describe a simplified fetch-decode-execute cycle using PC, MAR, MDR, CIR and the control unit. [6] (c) Explain one role of an interrupt and an interrupt service routine. [4]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

For an 8-bit signed representation, adding a ninth bit automatically gives the correct 8-bit answer.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.